

API BASE ENVIRONMENTS & REVERSE PROXIES

- Local Backend (Node.js Direct): <http://localhost:3000/api> [PORT 3000]
- Local Frontend Proxy (Vite Dev): <http://localhost:5500/api> [PORT 5500]
- Live Production (Vercel Cloud): <https://ithunt.vercel.app/api> [PRODUCTION]

1. System & Health Endpoints

GET

/api/health

Backend server & database connection health check (returns database: ithunt)

Local: <http://localhost:3000/api/health>

Live: <https://ithunt.vercel.app/api/health>

GET

/

API Engine root welcome banner and status

Local: <http://localhost:3000/>

Live: <https://ithunt.vercel.app/>

2. Admissions Management

GET

/api/admissions

Retrieve all candidate online admission applications

Local: <http://localhost:3000/api/admissions>

Live: <https://ithunt.vercel.app/api/admissions>

GET

/api/admissions/:id

Retrieve single admission record by Registration Number or ID

Local: <http://localhost:3000/api/admissions/:id>

Live: <https://ithunt.vercel.app/api/admissions/:id>

POST

/api/admissions

Submit new admission (auto-creates records in admissions, students, and users)

Local: <http://localhost:3000/api/admissions>

Live: <https://ithunt.vercel.app/api/admissions>

PATCH

/api/admissions/:id/status

Update candidate admission status (Confirmed / Pending / Verified)

Local: <http://localhost:3000/api/admissions/:id/status>

Live: <https://ithunt.vercel.app/api/admissions/:id/status>

DELETE

/api/admissions/:id

Delete admission application from database

Local: <http://localhost:3000/api/admissions/:id>

Live: <https://ithunt.vercel.app/api/admissions/:id>

3. Students & Academic Profiles

GET

/api/students

List enrolled students with optional filters (?course=&batch=&status=)

Local: <http://localhost:3000/api/students>

Live: <https://ithunt.vercel.app/api/students>

GET

/api/students/:id

Get student academic details by enrollment number or ID

Local: <http://localhost:3000/api/students/:id>

Live: <https://ithunt.vercel.app/api/students/:id>

POST

/api/students/register

Direct student registration into academic roster

Local: <http://localhost:3000/api/students/register>

Live: <https://ithunt.vercel.app/api/students/register>

PUT

/api/students/:id

Update student academic details and profile information

Local: <http://localhost:3000/api/students/:id>

Live: <https://ithunt.vercel.app/api/students/:id>

DELETE

/api/students/:id

Remove student profile from roster

Local: <http://localhost:3000/api/students/:id>

Live: <https://ithunt.vercel.app/api/students/:id>

4. Authentication & User Accounts

POST

/api/auth/login

Authenticate student or admin; returns JWT bearer token

Local: <http://localhost:3000/api/auth/login>

Live: <https://ithunt.vercel.app/api/auth/login>

GET

/api/auth/me

Get currently authenticated user details from token

Local: <http://localhost:3000/api/auth/me>

Live: <https://ithunt.vercel.app/api/auth/me>

GET

/api/auth/users

List all registered portal accounts (Requires Admin token)

Local: <http://localhost:3000/api/auth/users>

Live: <https://ithunt.vercel.app/api/auth/users>

PUT

/api/auth/users/:id

Update user account role, credentials, and profile

Local: <http://localhost:3000/api/auth/users/:id>

Live: <https://ithunt.vercel.app/api/auth/users/:id>

DELETE

/api/auth/users/:id

Delete user account

Local: <http://localhost:3000/api/auth/users/:id>

Live: <https://ithunt.vercel.app/api/auth/users/:id>

5. NIELIT 'O' & 'A' Level Projects

GET

/api/nielit-projects

List all NIELIT project guide applications and payment UTRs

Local: <http://localhost:3000/api/nielit-projects>

Live: <https://ithunt.vercel.app/api/nielit-projects>

GET

/api/nielit-projects/:id

Get single NIELIT project guide submission

Local: <http://localhost:3000/api/nielit-projects/:id>

Live: <https://ithunt.vercel.app/api/nielit-projects/:id>

POST

/api/nielit-projects

Submit project guide request and documentation

Local: <http://localhost:3000/api/nielit-projects>

Live: <https://ithunt.vercel.app/api/nielit-projects>

PUT

/api/nielit-projects/:id

Update NIELIT guide assignment, approval, or remarks

Local: <http://localhost:3000/api/nielit-projects/:id>

Live: <https://ithunt.vercel.app/api/nielit-projects/:id>

DELETE

/api/nielit-projects/:id

Delete NIELIT project application

Local: <http://localhost:3000/api/nielit-projects/:id>

Live: <https://ithunt.vercel.app/api/nielit-projects/:id>

6. Careers & Faculty Hiring

GET

/api/careers/applications

List faculty and instructor job applications

Local: <http://localhost:3000/api/careers/applications>

Live: <https://ithunt.vercel.app/api/careers/applications>

POST

/api/careers/apply

Submit instructor or faculty employment application

Local: <http://localhost:3000/api/careers/apply>

Live: <https://ithunt.vercel.app/api/careers/apply>

PATCH

/api/careers/applications/:id/status

Update applicant status (Shortlisted / Interviewed / Hired)

Local: <http://localhost:3000/api/careers/applications/:id/status>

Live: <https://ithunt.vercel.app/api/careers/applications/:id/status>

DELETE

/api/careers/applications/:id

Delete job application

Local: <http://localhost:3000/api/careers/applications/:id>

Live: <https://ithunt.vercel.app/api/careers/applications/:id>

7. Internships (3-Month & 6-Month)

GET

/api/internships/applications

List student software internship applications

Local: <http://localhost:3000/api/internships/applications>

Live: <https://ithunt.vercel.app/api/internships/applications>

POST

/api/internships/apply

Apply for software engineering internship

Local: <http://localhost:3000/api/internships/apply>

Live: <https://ithunt.vercel.app/api/internships/apply>

PATCH

/api/internships/applications/:id/status

Update internship progress or completion status

Local: <http://localhost:3000/api/internships/applications/:id/status>

Live: <https://ithunt.vercel.app/api/internships/applications/:id/status>

DELETE

/api/internships/applications/:id

Remove internship application

Local: <http://localhost:3000/api/internships/applications/:id>

Live: <https://ithunt.vercel.app/api/internships/applications/:id>

8. Event RSVPs (Campus Summits & Hackathons)

GET

/api/events/rsvps

List attendees registered for tech summits and hackathons

Local: <http://localhost:3000/api/events/rsvps>

Live: <https://ithunt.vercel.app/api/events/rsvps>

POST

/api/events/rsvp

Register attendance / RSVP for upcoming tech event

Local: <http://localhost:3000/api/events/rsvp>

Live: <https://ithunt.vercel.app/api/events/rsvp>

DELETE

/api/events/rsvps/:id

Cancel event reservation

Local: <http://localhost:3000/api/events/rsvps/:id>

Live: <https://ithunt.vercel.app/api/events/rsvps/:id>

9. Student Reviews & Ratings

GET

/api/reviews

List approved student reviews and star ratings

Local: <http://localhost:3000/api/reviews>

Live: <https://ithunt.vercel.app/api/reviews>

POST

/api/reviews

Submit student review and feedback

Local: <http://localhost:3000/api/reviews>

Live: <https://ithunt.vercel.app/api/reviews>

PATCH

/api/reviews/:id/approve

Admin approval for public display

Local: <http://localhost:3000/api/reviews/:id/approve>

Live: <https://ithunt.vercel.app/api/reviews/:id/approve>

DELETE

/api/reviews/:id

Delete review

Local: <http://localhost:3000/api/reviews/:id>

Live: <https://ithunt.vercel.app/api/reviews/:id>

10. Fees Ledger & Receipts

GET

/api/fees

View complete financial fee transactions ledger

Local: <http://localhost:3000/api/fees>

Live: <https://ithunt.vercel.app/api/fees>

GET

/api/fees/:id

Retrieve fee receipt details by receipt number

Local: <http://localhost:3000/api/fees/:id>

Live: <https://ithunt.vercel.app/api/fees/:id>

POST

/api/fees

Record new student fee payment and generate receipt

Local: <http://localhost:3000/api/fees>

Live: <https://ithunt.vercel.app/api/fees>

DELETE

/api/fees/:id

Delete fee payment entry

Local: <http://localhost:3000/api/fees/:id>

Live: <https://ithunt.vercel.app/api/fees/:id>

11. Certificates & Public Verification

GET

/api/certificates

List verified student certificates

Local: <http://localhost:3000/api/certificates>

Live: <https://ithunt.vercel.app/api/certificates>

GET

/api/certificates/verify/:certNo

Public certificate verification registry

Local: <http://localhost:3000/api/certificates/verify/:certNo>

Live: <https://ithunt.vercel.app/api/certificates/verify/:certNo>

POST

/api/certificates

Issue new verified student certificate

Local: <http://localhost:3000/api/certificates>

Live: <https://ithunt.vercel.app/api/certificates>

DELETE

/api/certificates/:id

Revoke certificate

Local: <http://localhost:3000/api/certificates/:id>

Live: <https://ithunt.vercel.app/api/certificates/:id>

12. Student Capstone Projects

GET

/api/projects

List capstone projects and live GitHub demo repositories

Local: <http://localhost:3000/api/projects>

Live: <https://ithunt.vercel.app/api/projects>

POST

/api/projects

Submit student project showcase

Local: <http://localhost:3000/api/projects>

Live: <https://ithunt.vercel.app/api/projects>

DELETE

/api/projects/:id

Remove project

Local: <http://localhost:3000/api/projects/:id>

Live: <https://ithunt.vercel.app/api/projects/:id>

13. Contact Inquiries & Admin Analytics

GET	/api/contact List customer course inquiries Local: http://localhost:3000/api/contact Live: https://ithunt.vercel.app/api/contact
POST	/api/contact Submit public course inquiry message Local: http://localhost:3000/api/contact Live: https://ithunt.vercel.app/api/contact
GET	/api/admin/stats Real-time aggregated counts across all 12 MongoDB collections Local: http://localhost:3000/api/admin/stats Live: https://ithunt.vercel.app/api/admin/stats
POST	/api/admin/firebase/sync-all Trigger 2-way cloud synchronization Local: http://localhost:3000/api/admin/firebase/sync-all Live: https://ithunt.vercel.app/api/admin/firebase/sync-all

TESTING & VERIFICATION COMMANDS (CURL)

```
curl -s http://localhost:3000/api/health # Check server & database status (ithunt)
curl -s http://localhost:3000/api/admissions # Retrieve active online admissions list
curl -s http://localhost:3000/api/certificates/verify/ITH-CERT-2026-001 # Public verification registry
curl -s http://localhost:3000/api/admin/stats # Real-time metrics across 12 collections
npm run db:verify # Node CLI verification tool
```